

NANYANG TECHNOLOGICAL UNIVERSITY
School of Social Sciences
HE3002 Macroeconomics III

LECTURE 3
Balance of Payments and the Current Account

READINGS

1. Olivier Blanchard *Macroeconomics*, 8th Edition (2021), Prentice Hall, Chapters 17 and 18.

THE BALANCE OF PAYMENTS

1. A country's international transactions are recorded in the **balance-of-payments**, BP .
2. BP has three main components:
 - (a) Current account, CA
 - (b) Financial account, FA , and
 - (c) Capital account, KA .
3. Therefore, BP is usually written as:

$$BP = CA + FA + KA \quad (1)$$

4. Since capital account balance is usually negligible, we are not discussing it here. Hence,

$$BP = CA + FA \quad (2)$$

5. The current account records (a) exports and imports of goods and services and (b) international receipts or payments of income.
6. Exports and income receipts enter with a plus (+) and imports and income payments enter with a minus (−).
7. For example, if a US resident buys a smartphone from South Korea for \$500, then the US current account goes down by \$500. This is because this transaction represents an import of goods worth \$500. Accordingly, the US current account records a negative entry of \$500.
8. The financial account keeps record of transactions in financial assets between residents and non-residents. Sales of assets to nonresidents represent an export of an asset and are given a positive sign (+) (financial inflows) in the financial account . Purchases of assets from nonresidents represent an import of a financial asset and enter the financial account with a negative sign (−) (financial outflows).

9. For example, in the case of the import of the smartphone, suppose the US resident pays for the phone with US currency, then this represents a sale (export) of US financial assets (currency) to a South Korean resident (Samsung) in the amount of \$500. Accordingly, the US financial account records a positive entry of \$500.
10. The smartphone example illustrates a fundamental principle of balance-of-payments accounting known as **double-entry bookkeeping**.
11. Each transaction enters the balance of payments twice, once with a positive sign and once with a negative sign.
12. An implication of the double-entry bookkeeping methodology is that any change in the current account must be reflected in an equivalent change in the country's financial account, that is, the current account equals the difference between a country's purchases of assets from foreigners and its sales of assets to them, which is the financial account preceded by a minus sign. This relationship is known as the **fundamental balance-of-payments identity**.
13. Because any international transactions automatically gives rise to two offsetting entries in the BoP, we have

$$\text{Current account balance} = - \text{Financial account balance} \quad (3)$$

14. The current account, on the other hand, consists of three components:

$$\text{Current account} = \text{Trade balance} + \text{Income balance} + \text{Net unilateral transfer} \quad (4)$$

15. The trade balance or balance on goods and services is the difference between exports and imports of goods and services. It consists of
 - (a) Merchandise trade balance or balance on goods, and
 - (b) Services balance.
16. The merchandise trade balance keeps record of net exports of goods.
17. The services balance keeps record of net receipts from items such as transportation, travel expenditures and legal assistance.
18. The income balance has
 - (a) Net investment income, and
 - (b) Net investment compensation to employees.
19. Net investment income is the difference between income receipts on US-owned assets abroad and income payments on foreign-owned assets in the US. It includes items such as international interest, dividend payments and earnings of domestically owned firms operating abroad.

20. Net investment compensation to employees includes

- (a) As positive entries: (1) earnings of US residents employed temporarily abroad, (2) earnings of US residents employed by foreign governments in US, and (3) earnings of US residents employed by international organizations in US.
- (b) As negative entries: (1) foreign workers who commute to work in US, (2) foreign students studying in US, (3) foreign professionals temporarily residing in US, and (4) foreign temporary workers in US.

21. Net unilateral transfers keeps record of the difference between gifts. One common item is private remittances.

22. The financial account has two main components as follows:

$$\text{Financial account} = \text{Increased in foreign-owned assets in the US} \quad (5)$$

$$- \text{Increased in US-owned assets abroad} \quad (6)$$

23. Note that foreign-owned assets in the US includes

- (a) US securities held by foreign residents
- (b) US currency held by foreign residents
- (c) US borrowing from foreign banks
- (d) Foreign direct investment in the US

24. US-owned assets abroad are

- (a) Foreign securities
- (b) US bank lending to foreigners
- (c) US direct investment abroad

25. An international transaction does not necessarily have to give rise to one entry in the current account and one entry in the financial account.

26. It can be the case that it gives rise to two offsetting entries in the financial account or two offsetting entries in the current account.

27. The example given earlier, namely, importing a smartphone and paying for it with cash, gives rise to one entry in the current account and one entry in the financial account.

28. International transactions that involve the exchange of financial assets generate two entries in the financial account and no entry in the current account. For example, if a US resident purchases

shares from Italy paying with dollars, then the financial account receives both a positive entry (the sale, or export, of dollars to Italy) and a negative entry (the purchase, or import, of equity shares from Italy).

29. It can also be the case that a transaction generates two offsetting entries in the current account and no entry in the financial account.
30. Suppose the US donates medications worth \$10 million to an African country afflicted by Malaria. This gift gives rise to two entries in the current account.
 - (a) The export of the malaria medication would be recorded in the merchandise trade balance, which is part of the current account, with +\$10 million.
 - (b) As the US does not receive an item of economic value in return, the offset is an entry of -\$10 million in net unilateral transfers, which is also part of the current account.
31. There is another type of transaction that merits separate discussion. This type of transaction is the purchase or sale of official reserve assets by central banks.
32. Official international reserves are foreign assets held by central banks as a cushion against national economic misfortune. At one time, official reserves consisted largely gold, but today central bank's reserves include substantial foreign financial assets.
33. Central banks often buy or sell international reserves in private asset markets to affect macroeconomic conditions in their economies. Official transactions of this type are called **official foreign exchange interventions**.

[Table 1: The US balance of payments in 2018 in billions of USD]

THE CURRENT ACCOUNT AND THE NET INTERNATIONAL INVESTMENT POSITION

1. One reason why the concept of current account balance is economically important is that it reflects a country's net borrowing needs. For example, in 2018 the US ran a current account deficit of \$488.5 billion. To pay for this deficit, the country must have either
 - (a) reduced part of its international asset position or
 - (b) increased its international liability position or
 - (c) both.
2. In this way, the current account is related to changes in a country's net international investment position (*NIIP*).
3. The term net international investment position is used to refer to a country's net foreign wealth, that is, the difference between the value of foreign assets owned by the country's residents and the value of the country's assets owned by foreign residents.
4. The net international investment position is a stock, while the current account is a flow.

[Figure 1: The US current account and net international investment position]

5. The US net international investment position was positive at the beginning of the sample. However, in the early 1980s, the US began running large current account deficits.
6. By 1989 these deficits had eroded the net foreign wealth of the US and she became a net debtor to the rest of the world for the first time since World War I.
7. The US current account deficits of the 1980s did not turn out to be temporary. As a consequence, by the end of the 1990s, the US had become the world's largest external debtor.
8. Moreover, the current account deficits continued to rise for twenty five years. Only shortly before the onset of the Great Recession of 2008, did the downward trend stop and current account deficits stabilized. In fact, they became smaller in magnitude.
9. By the end of 2018, the net international investment position of the US stood at -9.6 trillion dollars or -47 percent of GDP. This is a big number, and many economist wonder whether the observed downward trend in the net international investment position will be sustainable over time.
10. The concern stems from the fact that countries that accumulated large external debt to GDP ratios in the past, such as many Latin American countries in the 1980s, Southeast Asian countries in the 1990s, and more recently peripheral European countries, have experienced sudden reversals in international capital flows that were followed by costly financial and economic crises.

SAVING, INVESTMENT, AND THE CURRENT ACCOUNT BALANCE

1. Recall that the equilibrium in the goods market suggests that investment equals to saving, which is the sum of private saving and public saving.

2. We can now derive the corresponding condition for the open economy.

3. At equilibrium,

$$Y = C + I + G + (X - \frac{IM}{\epsilon}) \quad (7)$$

4. Rearrange gives:

$$Y - T - C = I + (G - T) + NX \quad (8)$$

where $NX = X - IM/\epsilon$.

5. Recall that, in an open economy, the total income of domestic residents is equal to domestic income, which is equal to domestic output, Y , plus net income from abroad, NI .

6. Add NI to both sides of the equation gives:

$$(Y + NI - T) - C = I + (G - T) + (NX + NI) \quad (9)$$

where $(Y + NI - T) - C$ is the private saving, S_p and $T - G$ is the public saving, S_g and $NX + NI$ is the current account, CA .

7. As a result,

$$S_p = I + (G - T) + CA \quad (10)$$

8. Or,

$$CA = S_p + (T - G) - I \quad (11)$$

9. And

$$CA = S_p + S_g - I \quad (12)$$

10. Given national saving, $S = S_p + S_g$, then:

$$CA = S - I \quad (13)$$

11. The current account balance is equal to national saving (private plus public) minus investment.

(a) A current account **surplus** implies that the country is saving **more than** it invests.

(b) A current account **deficit** implies that the country is saving **less than** it invests.

12. Note some of the things Equation (13) says:

- (a) An increase in investment must be reflected in either an increase in private or public saving or a deterioration in CA - a smaller CA surplus, or a larger CA deficit, depending on whether the current account is initially in surplus or deficit.
- (b) A deterioration in the government budget balance - either a smaller budget surplus or a larger budget deficit - must be reflected either in an increase in private saving or a decrease in investment, or a deterioration of the CA.
- (c) A country with a higher saving rate (private plus public) must have either a high investment or a large CA surplus.

WHO LENDS AND WHO BORROWS AROUND THE WORLD?

1. The large observed US current account deficits must be matched by current account surpluses of other countries with the US.
2. Over the past decade, an increasing fraction of the US current account deficit is accounted for by current account deficits with China.

[Figure 2: Cumulative Current Account Balances Around the World: 1980-2017]

3. The US current account with China as a fraction of the total US current account balance was about 20% in 1999 and has been increasing steadily, reaching a peak of 70% in 2009.